The 'hello world' of Monte Carlo

Part 1: The $e^+e^- \to q\bar{q}$ differential and total cross section

We will consider the process

$$e^+e^- \to \gamma^* \to q\bar{q}.$$

We work under the assumption that only the photon-exchange diagram is relevant. Furthermore, we assume that we are in the centre-of-mass (CM) frame of the colliding electrons (i.e. they are back-to-back) and assume negligible particle masses. The CM energy is denoted by $\sqrt{s}$. We denote the momenta of incoming electron-positron pair by p_1 and p_2, and those of the outgoing quarks by q_1 and q_2.

During the lecture, we saw that the matrix element is given by

$$iM_{e^+e^-\to q\bar{q}} = \frac{ie^2 Q_q}{(q_1+q_2)^2}\bar{u}(p_1)\gamma^\mu v(p_2)\,\bar{v}(q_2)\gamma_\mu u(q_1)\,,$$

with Q_q the charge of the quark, and e the electromagnetic coupling.

1. Using Dirac algebra, show that the squared matrix element is equal to $|M|^2 = e^4 Q_q^2 N_c(1+\cos^2\theta)$, where θ is the angle between the outgoing quark and the electron (in the CM frame), and the number of colours is denoted by N_c. Remember that all particles are taken to be massless. You will need

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\sigma\gamma^\rho] = 4\left(g^{\mu\nu}g^{\sigma\rho} - g^{\mu\sigma}g^{\nu\rho} + g^{\mu\rho}g^{\nu\sigma}\right).$$

2. Draw the distribution in θ of the squared matrix element from $\theta = 0$ to $\theta = \pi$.

3. Starting from the defining expression of the Lorentz-invariant phase space for two particles, i.e.

$$\int d\Phi_2 = \int \frac{d^3q_1}{2E_q(2\pi)^3}\frac{d^3q_2}{2E_{\bar{q}}(2\pi)^3}(2\pi)^4\delta(q_1^2)\delta(q_2^2)\delta^4(p_1+p_2-q_1-q_2)\,,$$

   show that

$$\int d\Phi_2 = \frac{1}{8\pi}\int d\cos\theta\,.$$

   Why are we allowed to integrate over the azimuthal angle ϕ?

Taken together, we find that the total cross section for $e^+e^- \to \gamma^* \to q\bar{q}$ becomes

$$\sigma(e^+e^-\to\gamma^*\to q\bar{q}) = \frac{4\pi\alpha_{\mathrm{EM}}^2}{3s}N_c\sum Q_q^2\theta(m_q^2 > s)\,.$$

These results will be used to write and validate our Monte Carlo.

Part 2: Our first event generator

Instead of computing the cross section analytically, we now want to set up a program that calculates it numerically. We will assume you work in python, but you are free to use any other language you are comfortable with. Random numbers between 0 and 1 may be obtained using

```
import random
r = random.uniform(0,1)
```

1. Using the fact that

$$\int_{x_{\min}}^{x} dx\, f(x) = r\int_{x_{\min}}^{x_{\max}} dx\, f(x)$$

   write a program returns a sample of x values that is distributed according to $f(x) = 1 + x^2$ with $x_{\min} = -1$ and $x_{\max} = 1$. Hint: one can use a bisection method to find the solution to an equation of the form $g(x) = 0$. Plot the resulting distribution in x. Does it agree with your expectation (part 1, item 2)? How many samples does one need to get a satisfactory distribution?

2. Write a program that returns a sample of x values distributed according to $f(x) = 1 + x^2$ using an accept-and-reject method. Which of (1) and (2) is faster?

3. Numerically compute the integral of
$$\int_{-1}^{1} dx(1+x^2)\,.$$

4. How can you compute the integral and at the same time obtain a set of x values distributed according to $f(x)$?

5. Defining now all relevant constants, which we set to $\alpha_{\rm EM} = 1/137$, $N_c = 3$, $Q_u = Q_c = 2/3$, $Q_d = Q_s = Q_b = -1/3$, plot the total cross section of $e^+e^- \to \gamma^* \to q\bar{q}$ in pb as a function of the scattering energy s for $\sqrt{s} = 1 - 100$ GeV.

You’ve now written an LO event generator that computes σ and $d\sigma/d\cos\theta$!

Part 3: Our first NLO event generator

At NLO we have to deal with the soft and collinear singularities that pop up. In this exercise, we will consider a toy model, and compute the NLO cross section in two ways: *subtraction* and *slicing*. This toy model only has one divergence, and hides some of the complexities one has to deal with when calculating an actual NLO correction to e.g. the $e^+e^- \to q\bar{q}$ process considered before.

Consider a system radiating off massless particles with energy x. At Born level, we do not have any emissions, and $x = 0$. At NLO we need to take into account loop corrections and real emission corrections, where the former is plagues by a virtual divergence, whereas the latter as a singularity when $x \to 0$. We define the following differential cross sections for the Born, Virtual and Real contributions:

$$\begin{aligned}
\left(\frac{d\sigma}{dx}\right)_{\rm B} &= B\delta(x)\,,\\
\left(\frac{d\sigma}{dx}\right)_{\rm V} &= \alpha\left(\frac{B}{2\epsilon} + V_f\right)\delta(x)\,,\\
\left(\frac{d\sigma}{dx}\right)_{\rm R} &= \alpha\frac{R(x)}{x}\,.
\end{aligned} \tag{1}$$

Here we have introduced a dimensional regulator ϵ, in addition to arbitary constants B and V_f, the coupling α, and a to-be-specified function $R(x)$. To compute the total cross section we need to evaluate

$$\sigma = \lim_{\epsilon\to 0}\int_0^1 dx\, x^{-2\epsilon}\left[\left(\frac{d\sigma}{dx}\right)_{\rm B} + \left(\frac{d\sigma}{dx}\right)_{\rm V} + \left(\frac{d\sigma}{dx}\right)_{\rm R}\right] \equiv \sigma_B + \sigma_V + \sigma_R\,. \tag{2}$$

Clearly, we cannot dump this on a computer as-is: there is a limit of $\epsilon \to 0$ to be taken care of.

1. Compute the contributions from the Born and virtual terms, $\sigma_B + \sigma_V$.

2. *Subtraction*: For the subtraction technique, we write the real contribution as
$$\sigma_R = \alpha\left[\int_0^1 dx\, x^{-2\epsilon}\frac{R(x) - R(0)}{x} + R(0)\int_0^1 dx\, x^{-1-2\epsilon}\right].$$
   (a) Evaluate the rightmost integral.
   (b) What must be the condition on $R(0)$ such that the divergence in the virtual contribution cancels?
   (c) What must be the condition on $R(x)$ such that the first integral is integrable in 4 dimensions ($\epsilon = 0$)?
   (d) Write an expression for the total cross section σ that can be evaluated directly in 4 dimensions.

3. *Slicing*: In slicing, we introduce a cutoff parameter $\delta \ll 1$ to write

$$\int_0^1 dx = \int_0^\delta dx + \int_\delta^1 dx\,.$$

   (a) Approximate $R(x) = R(0)$ when $x < \delta$. Compute the contribution from the region $x < \delta$ analytically, and expand in the limit that $\delta \to 0, \epsilon \to 0$ with $\delta \gg \epsilon$. What must be the condition on $R(0)$ to cancel the virtual divergence?

   (b) Write the contribution for $x > \delta$ in a form suitable for numerical integration.

4. *Monte Carlo implementation:* We now pick a simple function satisfying the required properties

$$R(x) = B(1 + x(1-x))\,.$$

With this we obtain a total cross section of

$$\sigma = B\left(1 + \alpha\left(\frac{V_f}{B} + \frac{1}{2}\right)\right).$$

   (a) Convince yourself of this answer in three ways: using Eq. (2) directly, using *subtraction* and using *slicing*.

   (b) Write a Monte Carlo program that computes σ using *subtraction* and *slicing*. Take $B = 2$, $V_f = 1$ and $\alpha = 0.1$ for the numerical evaluation of the cross section. Which method is more stable? Can you pick δ arbitrary small?